Sandesh Sampath Kumar

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Summary

Backend Engineer with 4 years building distributed systems and APIs at scale in Java, C#/.NET Core, and Python across AWS and Azure, scaling one enterprise platform to **8**× throughput and **10**× global transaction growth. Owns the full lifecycle from architecture decisions through on-call incident response, and now builds production agentic AI systems (RAG, MCP tool-calling, LLM evals).

Technical Skills

Languages: Java, C#, Python, TypeScript, JavaScript
Backend & APIs: Spring Boot, .NET Core, FastAPI, Node.js, REST & Versioned APIs, Microservices, Async I/O
Cloud & Infra: AWS (Lambda, API Gateway, Step Functions, SQS/SNS, DynamoDB), Azure, Kubernetes, Docker, Terraform, CI/CD
Agentic AI: LLM Evals, MCP Tool-Calling, RAG, Agent Orchestration, Prompt Engineering, Embeddings, Qdrant, AWS Bedrock
Data & Reliability: Postgres/PostgreSQL, SQL Server, MongoDB, Redis, Kinesis (KDS), Observability (Datadog), On-Call, TDD

Experience

Software Engineer | **Backend & AI** June 2026 – Nov 2026
First Help Financial *Durham, NC (Remote)*

- Developed a production **agentic AI** underwriting system (**RAG**, **MCP** tools via **FastMCP**, **Strands**, **AWS Bedrock**) that cut loan-version comparison and calculation review **70%** in UAT (10 to 3 min per application), with tool-boundary authorization and PII redaction.
- Owned design and delivery of backend services and APIs on **AWS** serverless, authoring **architecture decision records (ADRs)**.
- Operated services on **Linux** behind **AWS ALBs**, with **Terraform** IaC, CI/CD pipelines, and **Datadog** observability.
- Instrumented **trace-based LLM evals** (**CloudWatch**, **Datadog**) scoring intermediate retrieval and tool-call steps.
- Introduced an AI-assisted plan/execute workflow (**Cursor** with structured prompting), raising delivery speed **35%**.

Software Engineer Intern | **Backend, Distributed Systems** May 2025 – Dec 2025
First Help Financial *Boston, MA (Remote)*

- Implemented serverless services (**Lambda**, **Step Functions**, **API Gateway**, DocumentDB) for high-volume financial workflows.
- Built serverless **async processing** pipelines (**KDS**, **SQS/SNS**) with retry, timeout, and idempotency for **reliability**.
- Developed **internal Python SDKs** for logging, API templates, and fault-handling with **95%+** unit test coverage.
- Created a reproducible AWS-local dev environment (SAM + LocalStack), cutting dev and debug time **55%**.

Application Developer | **Backend & Distributed Systems** Sep 2022 – Aug 2024
Thoughtworks *Bengaluru, India*

- Re-architected an on-prem monolith into distributed **.NET Core** microservices on **Azure** (clean architecture, SOLID, design patterns), scaling an enterprise ordering platform to **8**× throughput and **10**× global transaction growth.
- Deployed production microservices to **Kubernetes** via **Terraform** and **Azure DevOps CI/CD**, secured with **OAuth 2.0** (**Entra ID**).
- Ran **on-call rotations**, driving incident response and root-cause analysis via **Application Insights** monitoring.
- Delivered **15+ features cross-functionally**, presented sprint demos to client **stakeholders**, and **mentored** juniors (**XP**, **TDD**).
- Cut client integration time **60%** with a self-service portal, enabling **$16M+** in revenue, and tuned SQL and pooling for **3**× speed.

Systems Engineer | **Java, Spring Boot, Angular** Aug 2021 – Aug 2022
Tata Consultancy Services *Bengaluru, India*

- Engineered **Java** backend services for enterprise banking (**Spring Boot**, **JPA/Hibernate**), speeding up APIs and queries **20%**.
- Integrated **Angular** UI components with Spring Boot APIs for end-to-end workflows, unit-tested with **JUnit** and **Mockito**.

Projects

Career Advisor Backend | Python, FastAPI, Qdrant, Ollama, RAG Sep 2025 – Oct 2025

- Built a self-hosted **RAG** pipeline (**FastAPI**, **Qdrant**, Ollama, Docker): ingestion, chunking, **embeddings**, vector search.
- Engineered deterministic ranking and **caching** so identical inputs produce reproducible output.

DreamChaserAI | React, Python, FastAPI, Google Gemini Mar 2026

- Shipped a live full-stack AI platform with **Gemini**-powered college matching and personalized document generation.
- Designed a hard-filter + LLM-tiering architecture with **JWT** auth and **MongoDB Atlas**.

Education

Northeastern University | **Boston, USA** Sep 2024 – May 2026
Master of Science in Computer Science *GPA: **4.0/4.0***

Jyothy Institute of Technology | **Bengaluru, India** Aug 2017 – Aug 2021
Bachelor of Engineering in Computer Science *GPA: **8.66/10***